

BA3093 Simulation and Risk Analysis

Group Project Guidelines

DEADLINES

Presentation (PPT): May 10, 2025 @23:59

Written Report: May 24, 2025 @23:59

1. Overview and Objectives

The goal of this group project is to apply **simulation and risk analysis techniques** to a real-world business, engineering, or operations problem. You will model uncertainty, analyse risk, and derive actionable insights that can support managerial decision-making.

This is a business-oriented course. Beyond technical execution, you are expected to clearly articulate **why** a particular method is appropriate, **how** to interpret the outputs, and **what** decisions or recommendations follow from your analysis. Analytical rigour and practical relevance are equally valued.

2. Technical Options

Each group must choose **one or both** of the following two technical streams for their project. Your entire analysis — method, modelling, and reporting — should be grounded in the chosen stream.

Option A — Monte Carlo Simulation

Design and implement a Monte Carlo simulation model to quantify uncertainty and estimate risk in a business or decision context. Your project should include all of the following elements:

- Identify and justify the probability distributions used to model each key uncertain input (e.g., demand, cost, return, project duration).
- Run a sufficient number of iterations (typically $\geq 1,000$) and present the resulting output distribution.
- Compute and interpret relevant risk metrics such as Value at Risk (VaR), Conditional Value at Risk (CVaR), probability of loss, or percentile estimates.
- Conduct a sensitivity analysis (e.g., tornado chart) to identify which inputs most strongly drive output uncertainty.
- Translate simulation results into concrete business recommendations.

Example application domains: Gaming development cost/duration risk, financial portfolio risk, supply-chain demand uncertainty, insurance claims modelling, capital budgeting under uncertainty.

Option B — Queuing Models

Analyse a real-world waiting-line system using queuing theory and, where appropriate, discrete-event simulation. Your project should include all of the following elements:

- Identify and justify the queuing model structure (e.g., M/M/1, M/M/c, M/G/1) based on observed or estimated arrival and service processes.
- Estimate the key input parameters (λ , μ , number of servers, queue discipline) from real or public data.
- Compute and interpret steady-state performance measures: average waiting time, queue length, server utilisation, and probability of waiting.
- Analyse system sensitivity to changes in arrival rate or capacity (e.g., what happens during peak hours or if a server is added).
- Translate queuing results into actionable recommendations for system design or resource allocation.

Example application domains: Bank teller or checkout lanes, hospital emergency triage, call-centre staffing, airport security, restaurant service operations, ride-hailing dispatch.

3. Group Composition

Groups should consist of 2 to 8 students. Groups will be formed and registered during the designated class session. Each member is expected to contribute meaningfully to both the technical work and the written report.

4. Report Structure

The written report must follow the structure below. Each section should demonstrate both technical competence and business reasoning.

Section	Content Requirements
Cover Page	Project title; group number; section number; all members' names and student IDs; course name; submission date.
1. Problem Introduction	Define the business or operational problem. State the research question and explain why simulation or queuing analysis is an appropriate method.
2. Data and Parameters	Describe data sources and how input parameters were estimated. Include a parameter table. Justify all distributional or structural assumptions.
3. Model Design	Present the model structure with sufficient detail for replication. For Monte Carlo: specify all random variables and their distributions. For queuing: specify the model type (e.g., M/M/c) and all parameters.

4. Results and Analysis	Report simulation outputs or queuing metrics. Use tables and charts. Interpret results in plain language with reference to the business problem.
5. Sensitivity Analysis	Identify which parameters have the greatest impact on outcomes. Discuss implications for decision-making.
6. Conclusion and Recommendations	Summarise findings. Propose at least one concrete, actionable recommendation. Discuss limitations and potential extensions.
References	APA format. Every claim must be supported by a citation.
Appendix (optional)	Supplementary figures, extended tables, or code listings.

5. Group Presentation

Each group will present their project to the class. Presentations are an opportunity to demonstrate your analytical reasoning and ability to communicate findings to a non-specialist audience.

Duration	15–20 minutes presentation + 5–10 minutes Q&A
Presentation order	Determined by Group number
Slides format	PPTX or PDF; must be submitted before presenting May 10, 2025 @23:59
Hard copy	Print one copy of your slides to hand to the instructor before you present
Q&A focus	Be prepared to justify your modelling assumptions and interpret outputs on the spot

6. Formatting Requirements

All formatting rules below are strictly enforced. Reports that do not comply may be penalized.

Page limit	Up to 25 pages (strict upper limit, excluding cover page, references, and appendix)
Page size	A4
Margins	2.54 cm on all sides
Line spacing	Double spacing
Font	Times New Roman, 12 pt
Language	English only
Citations	APA style; every claim supported by at least one citation

7. Submission Instructions

- Presentation slides (PPTX or PDF): due by May 10, 2025 @23:59.
- Written report (PDF) and all supporting code/files (as a single ZIP): due by May 24, 2025 @23:59.
- File naming convention: GroupNumber (e.g., “Group3.pdf”).
- Submit only soft copies through iSpace unless instructed otherwise.

9. Generative AI Policy

You are permitted to use generative AI tools for proofreading purposes only — that is, for revising or correcting grammar and language in your report and slides. If you use an AI tool for this purpose, you must acknowledge it explicitly on the cover page or in the abstract.

The following uses of AI are strictly prohibited: generating analysis, producing simulation outputs, text writing, creating charts or visualisations, and interpreting results. Any detected breach will be treated as academic misconduct and result in a mark of zero.

Frequently Asked Questions

What makes a strong project?

A strong project is one that clearly connects a real-world problem to an appropriate model, executes the analysis rigorously, and communicates findings in a way that a manager could act upon. Complexity for its own sake is not rewarded. Choose the method that best fits your problem and justify that choice.

Can we use software other than Excel?

Yes. You may use any software appropriate to your chosen technical stream — for example, Python, R, Arena, or AnyLogic. Regardless of tool, all code or model files must be submitted with your report (as a ZIP archive) so the teaching team can reproduce your results.

How much real data do we need?

You are encouraged to use real or publicly available data to estimate your model parameters. If real data are unavailable, you may use realistic assumptions drawn from published literature or industry benchmarks, provided you clearly state and justify each assumption in the report.

Can we combine both technical options?

YES. If necessary, you can implement the two techniques for comparison (Monte Carlo Simulation or Queuing Models). Note that blending the two methods superficially will not be awarded extra points.

How should we approach the sensitivity analysis?

For Monte Carlo projects, a tornado chart or one-at-a-time parameter sweep is recommended. For queuing projects, vary the arrival rate (λ), number of servers (c), service rate (μ) etc, and report the effect on key performance measures. In both cases, discuss what the sensitivity results mean for managerial decisions.

How will the Q&A be assessed?

All group members may be asked questions. You should be able to explain your modelling assumptions, defend your parameter choices, and interpret any output shown on your slides. Q&A performance reflects individual preparation and understanding.

Good luck — we look forward to seeing your work!